

FBE 459: FINANCIAL DERIVATIVES

Term: Fall 2026

Meetings: T & Th 4:00–5:50 pm **Room:** JFF 239

Instructor: Jason Roderick Donaldson **Office:** HOH 819

Office Hours: Tuesdays 9:30–10:30 am (please email ahead)

Email: jasonrdonaldson@gmail.com **Web:** Brightspace

Final exam: Th 12/10, 4:30–6:30 pm (per Registrar)

Course Description

To study the economic foundations and practical use of derivatives: forwards, futures, options, replication, and the Fundamental Theorem of Asset Pricing (FTAP), binomial models, Black–Scholes (from FTAP), the market price of risk, and case-based risk management.

Objectives

To price common derivatives, design hedging and speculative strategies, connect prices to risk premia, interpret market prices (term structures and volatility surfaces), and apply tools in cases.

Course Schedule

Lecture #	Date (Day)	Material
1	8/25 (T)	Introduction & "Great Trades" – Why derivatives are used; big-picture trades and lessons
2	8/27 (Th)	(No-Arb) Law of One Price – Replication, bounds, put–call parity; institutional detail
3	9/1 (T)	Forwards – Pricing, hedging, cash vs. carry, basis, and swaps
	9/3 (Th)	PS1 due + review
4	9/8 (T)	Storage & the Forward Curve – Storage costs, convenience yield, backwardation/contango
5	9/10 (Th)	Futures (Metallgesellschaft) – Rolling hedges, financing/liquidity risk, lessons
	9/15 (T)	No class
	9/17 (Th)	PS2 due + review
6	9/22 (T)	Introduction to Options – Calls/puts, convexity, moneyness, early exercise intuition
7	9/24 (Th)	Other Instruments & Corporate Securities – Option strategies; corporate securities and risk-shifting
8	9/29 (T)	One-Step Binomial (replication) – Static replication; state prices on the tree
	10/1 (Th)	PS3 due + review (binomial)
	10/6 (T)	Exam 1 (in class)
	10/8 (Th)	No class (Fall Recess)

9	10/13 (T)	Risk-Neutral Pricing & FTAP
		– Risk-neutral probabilities; pricing kernels; FTAP
10	10/15 (Th)	Market Price of Risk (CAPM)
		– Markowitz/CAPM; connecting Q and CAPM; betas of options
11	10/20 (T)	Multi-Step Binomial
		– Backward induction; time aggregation; path dependence
	10/22 (Th)	PS4 due + review
12	10/27 (T)	Infinitely Repeated Tree
		– Limit of the binomial; log-normal link
13	10/29 (Th)	Black–Scholes (derivation from FTAP)
		– Assumptions; economic interpretation; applications
14	11/3 (T)	The Greeks
		– Δ , Γ , vega, Θ , ρ ; hedging with Greeks
15	11/5 (Th)	Volatility I
		– Implied volatility, VIX, and the volatility surface
	11/10 (T)	PS5 due + review
	11/12 (Th)	Exam 2 (in class)
16	11/17 (T)	Volatility II
		– Trading volatility; gamma/theta; fat tails and skew

17	11/19 (Th)	Big Game Case
		– Synthesis: pricing, hedging, governance, incentives
	11/24 (T)	PS6 due + review
	11/26 (Th)	No class (Thanksgiving)
18	12/1 (T)	Further Applications
		– E.g., real options
	12/3 (Th)	Final Review

Materials

Slides, problem sets, and cases will be posted on Brightspace. Recommended: Hull, *Options, Futures, and Other Derivatives*, or McDonald, *Derivatives Markets*.

Assessment

Participation (5%); Problem Sets & Cases (10%); Exam 1 (25%); Exam 2 (25%); Final (35%).

Exams are closed-book; calculators allowed.

USC Boilerplate

Use of AI. Use AI tools only when explicitly permitted; always cite sources.

Technology Requirements. You are responsible for ensuring that you have the necessary computer equipment and reliable internet access to complete this course. This course requires Microsoft Excel and access to Brightspace. Marshall has site licenses for a variety of software that students can access free of charge. A list of available software is located here. You are invited to explore what lab or loaner options exist. Contact the Marshall HelpDesk (213-740-3000 or HelpDesk@marshall.usc.edu) if you need assistance.

Open Expression and Respect for All. An important goal of the educational experience at USC Marshall is to be exposed to and discuss diverse, thought-provoking, and sometimes controversial ideas that challenge one's beliefs. In this course we will support the values articulated in the USC Marshall Open Expression Statement (<https://www.marshall.usc.edu/open-expression-statement>).

Recording. No recording or distribution without instructor permission.

Academic Integrity

The University of Southern California is foremost a learning community committed to fostering successful scholars and researchers dedicated to the pursuit of knowledge and the transmission of ideas. Academic misconduct is contrary to this fundamental mission and includes any act of dishonesty in the submission of academic work (either in draft or final form), as well as cheating, plagiarism, fabrication (e.g., falsifying data), knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage. Students are expected to uphold the highest standards of academic integrity in all coursework. This course follows the expectations for academic integrity as stated in the USC Student Handbook. All students are expected to submit assignments that are original work and prepared specifically for the course/section in this academic term. Students may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s). Students suspected of academic misconduct will be reported to the Office of Academic Integrity.

Academic dishonesty has a far-reaching impact and is considered a serious offense against the university. Violations will result in a grade penalty, such as a failing grade on the assignment or in the course, and disciplinary action from the university, such as suspension or expulsion. For more information about academic integrity see the Student Handbook, the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct. Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment or what information requires citation and/or attribution.

Instructors determine whether and how generative AI tools (such as ChatGPT, Copilot, Claude, Gemini, and similar applications) and other external resources (such as English-language translation tools) may be used in this course. Expectations may vary from assignment to assignment. This includes AI tools that record, transcribe, or summarize class sessions, which may require express instructor permission. Review the AI use guidance provided carefully, and direct any questions to your instructor before submitting work.

Students seeking general guidance may consult the following resources: USC Student Handbook; Generative AI General Policy; Practical Tips for Using AI in Academic Work; and Office of Academic Integrity.

Revised July 2026

Statement on University Academic and Support Systems

Students and Disability Accommodations

USC welcomes students with disabilities into all of the University's educational programs. The Office

of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Student Financial Aid and Satisfactory Academic Progress

To be eligible for certain kinds of financial aid, students are required to maintain Satisfactory Academic Progress (SAP) toward their degree objectives. Visit the Financial Aid Office webpage for undergraduate- and graduate-level SAP eligibility requirements and the appeals process.

Support Systems

Counseling and Mental Health – (213) 740-9355 – 24/7 on call. Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention. <https://sites.usc.edu/counselingandmentalhealth/>

988 Suicide and Crisis Lifeline – 988 for both calls and text messages – 24/7 on call. The 988 Suicide and Crisis Lifeline provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. <http://988lifeline.org/>

CARE-SC: Confidential Advocacy, Resources, and Education Support Center – (213) 740-9355(WELL) – 24/7/365 on call. Confidential advocates, prevention educators, and professional counseling teams work to promote a universal culture of consent, and prevent and respond to gender- and power-based harm. Services available to all USC students at no cost. <https://sites.usc.edu/clientservices/>

Office of Civil Rights Compliance – (213) 740-5086. Information about how to get help or help someone affected by harassment, discrimination, retaliation on the basis of a protected characteristic, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants. <http://eeotix.usc.edu/>

USC Campus Support and Intervention – (213) 740-0411, uscsupport@usc.edu. Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student. <http://campussupport.usc.edu/>

USC Emergency Information. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible. <https://emergency.usc.edu/>

USC Department of Public Safety. For 24 hour emergency assistance or to report a crime: UPC: (213) 740-4321, HSC: (323) 442-1000. For 24 hour non-emergency assistance or information: UPC: (213) 740-6000, HSC: (323) 442-1200. <https://dps.usc.edu/>

Office of the Ombuds – (213) 821-9556 (UPC) / (323) 442-0382 (HSC). A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern. <http://ombuds.usc.edu/>

Occupational Therapy Faculty Practice – (323) 442-2850 or otfp@med.usc.edu. Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance. <http://chan.usc.edu/patient-care/faculty-practice>

The **Office of Sustainability** promotes numerous ways for all Trojans to study, work, and live sustainably at USC. Students can access tools, resources, and opportunities on their website and in the Sustainability Hub in the Student Union Building, Suite 101. <https://sustainability.usc.edu/>

USC Report & Response provides a front door to reporting concerns, including an ability to report anonymously, which violate university policies or go against our Unifying Values and helps connect you with the support you need. The website provides a list of confidential resources, supporting offices, and policies that are available to our university community. The Resources page includes information regarding the university's response in the event of immigration enforcement activity on campus or involving a student or employee. Additional questions may be directed to the Office of Professionalism

and Ethics (OPE) and ope@usc.edu or in reviewing the Office of Professionalism and Ethics website.
<https://report.usc.edu/>

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